Owusu Kantankah

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Education

University of Wisconsin–Madison — Madison, WI
Bachelor of Science in Computer Science and Data Science — Expected May 2028

Experience

LASER Research Program, UW–Madison — Madison, WI
Software Engineer Intern — May 2026 – Aug. 2026

- Improved reinforcement-learning training throughput **2.3x** (0.884s → 0.384s per episode) using batched PyTorch operations, `torch.compile`, and CUDA graphs.
- Built a PyTorch/CUDA multi-agent system supporting **100 concurrent agents** across three interchangeable policy architectures.
- Enabled real-time browser inference and instanced rendering for 100 agents by porting trained policies to TensorFlow.js and integrating them into a Three.js client.
- Earned the L&S **Rising Researcher Award**, an academic-year return offer, and selection as a Computer Graphics TA.

Space Science and Engineering Center (SSEC) — Madison, WI
Undergraduate Data Engineering Researcher — Sep. 2025 – May 2026

- Built a Python/NumPy pipeline processing **10 GB+ of NetCDF telemetry** from JAXA's Akatsuki orbiter, reducing analysis runtime by 35%.
- Analyzed 1440×720 Venusian atmospheric grids, identifying a ~195 K polar vortex and 20 K equator-to-pole thermal gradient; presented findings at UW–Madison's Undergraduate Research Symposium.

StudioNil — Remote
Founder & Software Engineer — Mar. 2023 – Aug. 2025

- Founded and shipped a multiplayer game that reached **10,000 monthly active users within six weeks**.
- Architected client-server infrastructure supporting **100 concurrent players per instance at a 0.02% crash rate**.
- Built a raycast-based 3D hitbox system, improving collision accuracy by **1.8x** over Roblox's native `.Touched` API.

Projects

Godot Game Engine (Open Source) | *C++, SCons, Git* | GitHub

- Landed an **upstream C++ fix** for a Godot 4.7 animation regression in an engine with ~2M downloads per release.
- Diagnosed the regression with `git bisect` and transform-pipeline tracing, identifying incorrect matrix composition in non-uniform TRS animation and correcting local-space basis scaling.

3D Drone Simulation | *Three.js, GLSL, JavaScript* | GitHub

- Built a real-time boids-based swarm simulator for **1,000 autonomous agents** using Three.js and GLSL.
- Collapsed 1,000 draw calls into one instanced render, increasing frame rate from **80 to 133 FPS**.
- Replaced $O(n^2)$ pairwise neighbor search with a 27-cell uniform spatial hash, reducing queries to near-linear time.

Roblox Trade Intelligence System | *Python, Flask, SQLite, AWS* | GitHub

- Built a fault-tolerant pipeline using Gemini vision + text validation to parse trade proofs and infer trades from reciprocal asset-ownership transfers.
- Deployed on AWS Lightsail; indexed **100,000+ traders and 2,562 items**; linked reconstructed transactions to proofs for market-value estimates with **98.5% image/text agreement**.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, C, Java, SQL, GLSL, HTML/CSS
Frameworks/Libraries: PyTorch, React, Three.js, TensorFlow.js, Flask, NumPy, pandas, Tailwind
Tools: Git, Linux, CUDA, Docker, AWS, CMake, SQLite
Coursework: High Performance Computing (Graduate), Data Structures & Algorithms, Computer Graphics, Databases